AEGIS BANK

AI Risk Management Framework

Enterprise AI Decision & Knowledge Platform for Modern Banking

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Document Information

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Confidentiality Notice

This document defines the framework used by Aegis Bank to identify, assess, monitor, and mitigate risks associated with Artificial Intelligence systems.

The purpose of this framework is to ensure AI systems operate safely, transparently, securely, and responsibly within banking environments.

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1. Purpose

The purpose of this framework is to establish a structured approach for managing risks associated with AI technologies used at Aegis Bank.

This framework ensures:

- Reliable AI operations.
- Protection of customer information.
- Responsible AI usage.

- Regulatory compliance.
- Continuous improvement.

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2. Scope

This framework applies to:

- AI engineers.
- Data scientists.
- Security teams.
- Compliance teams.
- Software developers.
- Business users.

It covers:

- Large Language Models.
- RAG systems.
- AI assistants.
- Machine learning models.
- Automated decision-support systems.

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3. AI Risk Management Principles

Aegis Bank follows these principles:

Risk Prevention

Identify risks before deploying AI systems.

Transparency

AI decisions must be explainable.

Human Oversight

Critical decisions require human validation.

Continuous Monitoring

AI systems must be reviewed throughout their lifecycle.

Security First

AI systems must protect sensitive information.

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4. AI Risk Governance Structure

Aegis Bank maintains clear AI risk ownership.

Chief AI Officer (CAIO)

Responsibilities:

- AI strategy.
- Governance approval.
- Risk oversight.

AI Governance Team

Responsibilities:

- Risk assessments.
- Policy management.
- AI reviews.

Security Team

Responsibilities:

- AI security testing.
- Threat management.
- Data protection.

Compliance Team

Responsibilities:

- Regulatory alignment.
- Audit support.

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5. AI Risk Identification

Potential AI risks include:

Technical Risks

Examples:

- Model failure.
- Incorrect predictions.
- System downtime.

Security Risks

Examples:

- Prompt injection.
- Data leakage.
- Unauthorized access.

Data Risks

Examples:

- Incorrect documents.
- Poor-quality training data.
- Missing information.

Operational Risks

Examples:

- Poor user adoption.
- Incorrect AI usage.
- Lack of monitoring.

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6. AI Risk Assessment Process

Every AI system follows a risk assessment process.

Risk Identification

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Risk Analysis

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Risk Evaluation

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Risk Mitigation

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Continuous Monitoring

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7. Risk Classification Framework

Aegis Bank classifies AI risks based on impact.

Risk Level	Description
Low	Minimal business impact
Medium	Requires monitoring
High	Requires mitigation plan
Critical	Immediate action required

Example:

Risk	Level
Slow AI Response	Low
Incorrect Policy Answer	Medium
Customer Data Exposure	High
Unauthorized AI Access	Critical

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8. Model Risk Management

AI models must be evaluated before production use.

Evaluation includes:

- Accuracy testing.
- Performance testing.

- Security review.
- Bias evaluation.

Model Lifecycle

Development

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Testing

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Validation

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Approval

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Deployment

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Monitoring

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Retirement

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9. Hallucination Risk Management

Hallucination occurs when AI generates unsupported information.

Aegis AI OS reduces hallucination through:

- RAG architecture.
- Source verification.
- Prompt controls.
- Response evaluation.

Hallucination Control Flow

User Question

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Document Retrieval

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Relevant Context

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LLM Response

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Source Validation

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Confidence Score

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10. Data Risk Management

AI systems depend on reliable data.

Controls include:

- Document approval.
- Data validation.
- Access control.
- Data classification.

Data must be:

- Accurate.
- Complete.
- Secure.
- Updated.

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11. Security Risk Management

AI security controls include:

- Authentication.

- Authorization.
- Encryption.
- Monitoring.
- Threat detection.

Protected areas:

- AI prompts.
- Model configurations.
- Internal documents.
- Customer information.

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12. Bias & Fairness Management

Aegis Bank ensures AI systems provide fair results.

Controls:

- Testing datasets.
- Output analysis.
- Human review.
- Continuous evaluation.

Example:

Loan-related AI recommendations must not unfairly discriminate against customers.

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13. AI Compliance Controls

AI systems must follow:

- Internal banking policies.
- Data protection requirements.

- Security standards.
- Regulatory expectations.

Compliance activities:

- Regular reviews.
- Documentation.
- Audit records.
- Risk reporting.

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14. AI Monitoring Framework

Aegis AI OS continuously monitors:

Metric	Purpose
Accuracy	Response quality
Confidence Score	Reliability
Response Time	Performance
Error Rate	Failure detection
User Activity	Security tracking

Tools:

- LangFuse.
- Monitoring dashboards.
- Audit logs.

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15. Incident Management

AI incidents follow a structured process.

Detection

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Assessment

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Containment

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Resolution

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Review

Examples:

- Incorrect AI answer.
- Data leakage.
- Security attack.
- Model failure.

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16. Practical Examples

Example 1 — AI Gives Incorrect Policy Answer

Situation:

AI provides outdated information.

Response:

1. Review LangFuse trace.
2. Identify source document.
3. Update knowledge base.
4. Re-test AI response.

Example 2 — Unauthorized AI Access

Situation:

User attempts restricted document access.

Response:

1. Request blocked.
2. Security alert created.
3. Access reviewed.

Example 3 — RAG Document Error

Situation:

Wrong document retrieved.

Response:

1. Analyze retrieval logs.
2. Update document metadata.
3. Improve retrieval settings.

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17. Frequently Asked Questions (FAQs)

Q1. Why does Aegis Bank need AI risk management?

To ensure AI systems remain secure, accurate, and trustworthy.

Q2. How does Aegis reduce AI hallucination?

Through RAG, source validation, monitoring, and evaluation.

Q3. Can AI operate without human review?

No. Critical banking decisions require human oversight.

Q4. How are AI risks measured?

Through risk assessment, impact analysis, and monitoring metrics.

Q5. Who manages AI risks?

AI Governance, Security, Compliance, and Engineering teams.

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18. Revision History

Version	Date	Description	Author
v1.0	April 05, 2026	Initial AI Risk Management Framework Release	AI Governance Team
v1.1	May 30, 2026	Added Model Risk and Hallucination Controls	AI Architecture Team
v1.2	**July 24, 2026**	**Enhanced AI Risk Assessment, Compliance Controls, and Enterprise Monitoring Framework**	**Artificial Intelligence Department**

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End of Document

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